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Machine learning–based prediction of clinical mastitis in dairy cows: A comparative analysis of 9 algorithms using production and management data

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ABSTRACT

Mastitis represents one of the most formidable challenges in modern dairy farming, posing significant threats to individual cow health and causing substantial economic losses throughout the dairy production chain. Traditional disease diagnosis methods are often reactive and costly, creating an urgent need for advanced predictive technologies. To address these issues, we proposed a novel machine learning–based mastitis prediction system that breaks through conventional diagnostic paradigms by deeply integrating data science with veterinary medicine. We analyzed 177,493 dairy cow records from a large-scale commercial dairy farm in China, implementing 9 distinct machine learning algorithms for model development and evaluation: random forest, multilayer perceptron, support vector machine, decision tree, gradient boosting, adaptive boosting (AdaBoost), linear discriminant analysis, logistic regression, and naive Bayes. The dataset included comprehensive production metrics, physiological parameters, and management variables, with models trained on both standardized and nonstandardized datasets using rigorous cross-validation techniques. Random forest demonstrated superior predictive performance across all evaluation approaches. On synthetic minority oversampling technique (SMOTE)-balanced cross-validation, random forest achieved an F_1 score of 0.804 and an area under the receiver operating characteristic curve (AUC) of 0.884 (95% CI: 0.883–0.885). When evaluated using the methodologically rigorous approach with SMOTE applied within each cross-validation fold, performance metrics had an F_1 score of 0.542 and an AUC of 0.773, closely matching the held-out test-set results (F_1 score: 0.558, AUC: 0.785). This consistency between cross-validation and test-set performance validates our evaluation methodology and demonstrates robust model generalization under realistic class imbalance conditions. Feature importance analysis revealed month_age as the most critical predictor (relative importance: 0.165), followed by milk_yield (0.138), protein_percentage (0.138), and fat_percentage (0.135). The Z-standardization consistently enhanced model performance across all algorithms, with random forest maintaining optimal calibration between predicted probabilities and actual outcomes. Leveraging the predictive model, we quantified key risk factors and developed an early warning system capable of identifying high-risk animals, providing a robust foundation for precision dairy farming and improved mastitis management strategies. This research successfully develops a data-driven technological solution that substantially reduces disease spread risk and economic losses while driving the transformation of dairy farming toward digitalization and precision management.

Key words: cow, mastitis, machine learning, random forest, veterinary diagnostics

INTRODUCTION

Mastitis is one of the most devastating diseases in dairy farming worldwide, causing 22 billion dollars in annual losses globally through reduced milk yield, quality degradation, and increased treatment costs (Rasmussen et al., 2024). Clinical mastitis affects ~38% of US dairy cows annually, while subclinical mastitis prevalence ranges from 5% to 75% depending on management practices (Mramba and Mohamed, 2024; Jiang et al., 2025). Beyond immediate economic impacts, recurrent infections often require early culling, compromising genetic improvement programs and increasing herd turnover rates (Cha et al., 2010).

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Traditional mastitis diagnosis methods face numerous challenges in modern dairy farming. Conventional SCC testing is typically performed monthly, making this low-frequency monitoring unable to capture early disease signs, resulting in missed optimal intervention opportunities (Schukken et al., 2003; Dufour and Dohoo, 2013). High costs of detection equipment and specialized personnel make comprehensive monitoring economically unfeasible for many operations (Steeneveld et al., 2008). Most critically, traditional methods are predominantly reactive, diagnosing diseases only after clinical symptoms appear rather than providing proactive predictive capabilities (Eckelkamp, 2019).

The advent of precision dairy farming and machine learning offers transformative potential for mastitis prevention. Machine learning algorithms can systematically analyze daily production indicators and identify potential risk patterns before clinical symptoms appear, achieving true early warning capabilities (Hogeveen et al., 2010). Recent studies have demonstrated promising applications across various approaches. Ebrahimi et al. (2019) achieved over 90% sensitivity using 7 machine learning algorithms with automated monitoring data. Hyde et al. (2020) employed random forest and gradient boosting models to predict mastitis infection patterns, demonstrating superior performance compared to traditional statistical methods. Fadul-Pacheco et al. (2021) explored various machine learning algorithms for early mastitis prediction, highlighting the importance of feature selection and data preprocessing in model performance.

Machine learning approaches offer several advantages over traditional diagnostic methods. Unlike traditional single-indicator approaches, machine learning models can simultaneously consider multiple dimensions, including milk yield, milk composition, physiological status, and environmental factors, capturing complex nonlinear relationships to construct more comprehensive and accurate prediction frameworks (Norberg et al., 2004; Kamphuis et al., 2013). Recent advances have further expanded the scope and sophistication of machine learning applications in dairy farming. Satoła and Satoła (2024) compared ensemble machine learning models (bagging, boosting, stacking) against single algorithms for subclinical mastitis prediction in Polish Holstein-Friesian cows, demonstrating that support vector machines achieved the highest F_1 score (0.760–0.790) across multiple herds. Tian et al. (2024) developed clinical mastitis prediction models using milk yield, rumination time, and electrical conductivity data from Chinese dairy farms, with multilayer neural networks and random forest showing superior performance on standardized datasets. Thompson et al. (2023) implemented machine learning for predicting SCC status postcalving using routine UK milk recording data, emphasizing the critical importance of dry period management. The integration of time series analysis has also gained prominence, with Guo et al. (2025) exploring eXtreme Gradient Boosting (XGBoost) algorithms combined with SHapley Additive exPlanations (SHAP) value interpretation for mastitis prediction, achieving enhanced model interpretability alongside high accuracy. These algorithm-driven systems can efficiently process massive datasets without human resource limitations, providing feasible whole-herd monitoring solutions (Banhazi et al., 2012).

Despite these advances, current research faces several critical limitations that hinder widespread practical implementation. First, most studies utilize extensive feature sets requiring specialized measurements and equipment (pH meters, electrical conductivity sensors, climate monitoring stations) that may not be available in all dairy operations, particularly in resource-constrained environments. Second, many models are developed and validated on relatively small datasets or single-farm scenarios, limiting their generalizability across different farming systems and geographic regions (Miekley et al., 2013). Third, there is insufficient systematic evaluation of whether simpler, more accessible feature sets can achieve comparable or superior performance to complex approaches, with no previous study demonstrating that streamlined methodologies using standard farm data can outperform specialized measurement approaches. Finally, feature importance analysis and model interpretability remain underexplored, hindering the translation of research findings into actionable management strategies (Steeneveld et al., 2010).

To address these limitations, this study develops a streamlined, scalable machine learning approach for mastitis prediction using routine farm data. Our research makes several distinct contributions: (1) systematic evaluation of 9 machine learning algorithms on the largest single-farm dataset reported to date (177,493 records from 28,030 animals—approximately 45 times larger than previous studies), (2) implementation of random forest-based feature selection to identify optimal predictive variables from standard milk recording data, (3) comprehensive methodological innovations including animal ID–based data splitting to prevent data leakage and synthetic minority oversampling technique (**SMOTE**) application for class imbalance, and (4) detailed model calibration analysis for practical clinical decision-making. Unlike previous approaches requiring specialized equipment and measurements, our methodology uses only variables routinely collected during standard milk recording procedures, maximizing accessibility and cost-effectiveness while demonstrating superior predictive performance.

The organization of this article is structured as follows: the Materials and Methods section introduces the

dataset and proposed methodologies, the Results section presents the results, the Discussion section provides a comprehensive discussion, and the Conclusions section concludes the study and offers perspectives for future research.

MATERIALS AND METHODS

Study Design and Data Collection

This retrospective cohort study was conducted using comprehensive production records from a large-scale commercial dairy farm in Jiangsu Province, China. The study analyzed data collected over multiple lactation cycles from a single commercial dairy operation from January 2018 to December 2023, encompassing a total of 177,493 individual cow records from 28,030 unique dairy cows. All procedures were conducted in accordance with animal welfare guidelines and approved by the institutional animal care and use committee.

The dairy farm operated under standardized management protocols with automated data collection systems integrated into daily operations. Cows were housed in freestall barns with automated milking systems that recorded production parameters at each milking session. The herd consisted primarily of Holstein-Friesian cattle maintained under consistent feeding, housing, and health management protocols throughout the study period. Data were systematically collected from the farm's management information system, which integrated records from automated milking equipment, monthly milk testing services, and routine health monitoring programs conducted by trained farm personnel.

Study Population Characteristics

The study population included all lactating dairy cows present on the farm during the observation period. After applying inclusion criteria for complete production and health records, minimum SCC testing, age between 20 to150 mo, and at least 2 consecutive monthly test records, 28,030 individual cows contributed 177,493 test records to the final analysis. The study population demonstrated typical dairy herd demographics (as shown in Figure 1), with a mean age of 43.0 mo and predominance of first-lactation (36.5%) and second-lactation (35.0%) cows. Overall, 32.6% of cows never experienced mastitis during the observation period, while 67.4% had at least one positive test result based on SCS ≥4. The mastitis rate distribution represents the proportion of positive SCC tests relative to total tests per individual cow, with the majority having zero mastitis rate, while others exhibited rates from 0.1 to 1.0, reflecting intermittent subclinical and chronic infection patterns, respectively. Testing frequency averaged 6.3 records per cow, reflecting typical monthly sampling intervals.

Mastitis Definition and Classification

Clinical and subclinical mastitis cases were identified using SCC as the primary diagnostic criterion, following established international standards for dairy herd health monitoring. Mastitis was defined as SCC ≥200,000 cells/mL or SCS ≥4, consistent with widely accepted thresholds in veterinary practice and research protocols (Bonestroo et al., 2022a; Tian et al., 2024). This threshold was selected based on its demonstrated correlation with intramammary infection status and its widespread acceptance in dairy industry practice for distinguishing infected from noninfected quarters.

Records were classified into binary categories based on the SCS threshold: mastitis-positive (SCS ≥4) and mastitis-negative (SCS <4). The overall dataset contained 52,098 mastitis-positive records (29.3% of total tests) and 125,395 mastitis-negative records (70.7% of total tests), representing a moderate prevalence typical of commercial dairy herds. Individual cow mastitis status was further characterized by calculating the mastitis rate as the proportion of positive tests relative to total tests per cow, providing a more comprehensive measure of infection burden over time than single-point prevalence estimates.

Variable Collection and Feature Engineering

A comprehensive set of variables was systematically extracted from the farm's database, encompassing production performance, physiological characteristics, management factors, and temporal variables. The initial feature set included over 50 potential predictors covering various aspects of cow health, production, and management history. Unlike previous studies that utilized extensive feature sets including specialized measurements (Bobbo et al., 2023), we focused on variables routinely collected during standard milk recording procedures to maximize practical applicability. Our approach prioritizes accessibility and cost-effectiveness while maintaining predictive accuracy, addressing a key limitation in current mastitis prediction research where specialized equipment requirements may limit widespread adoption. Production variables comprised daily milk yield measured in kilograms per day, milk fat percentage, milk protein percentage, fat-to-protein ratio, corrected milk yield calculated as 305-d equivalent, peak milk yield and corresponding day of peak production, and lactation persistency indices calculated using standard dairy industry formulas.

Physiological variables captured cow age in months at the time of testing, parity number representing lactation

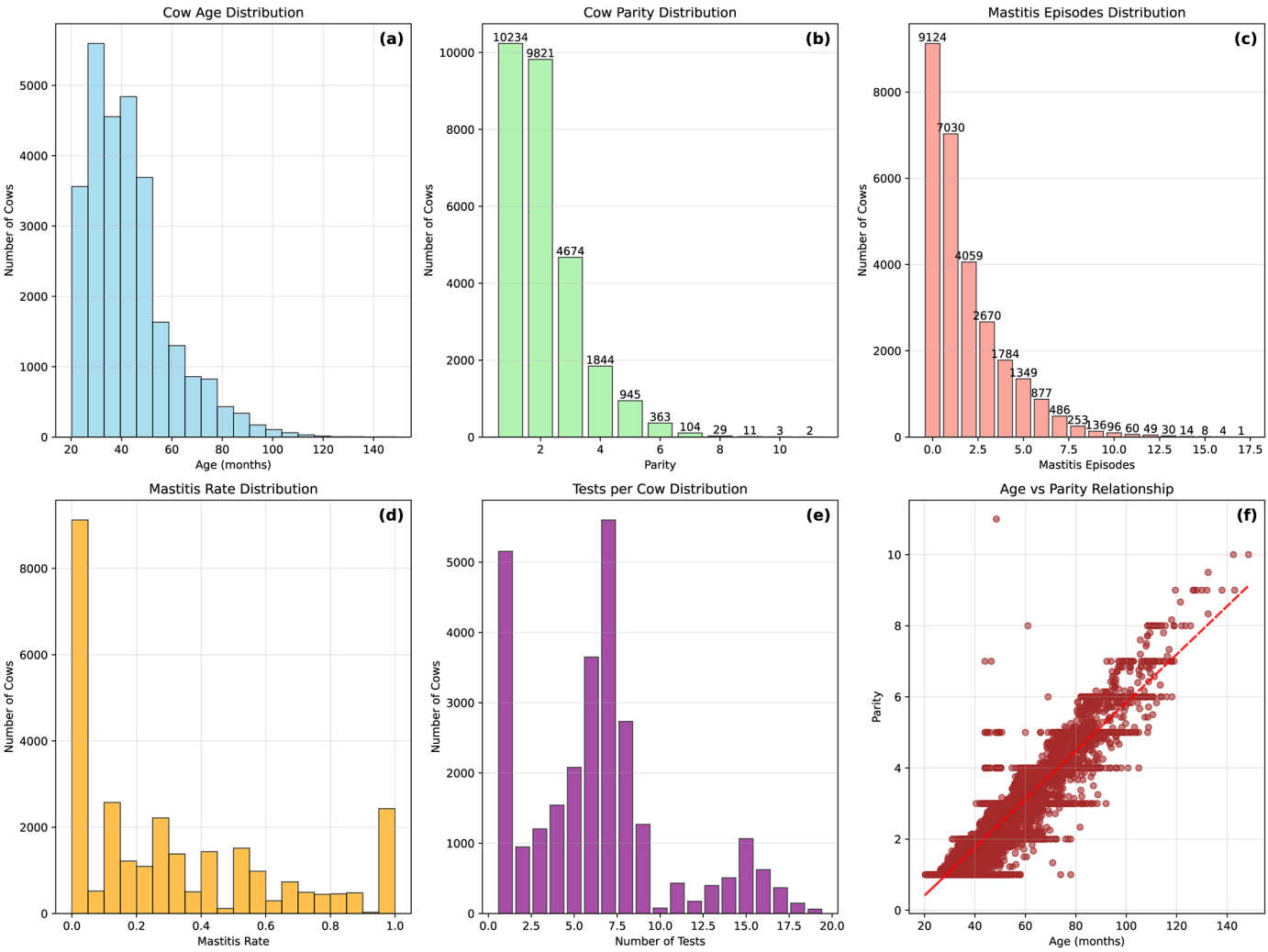


Figure 1. Descriptive characteristics of the study population. (a) Age distribution of dairy cows showing concentration in younger age groups; (b) parity distribution indicating predominance of first- and second-lactation animals; (c) distribution of mastitis episodes per cow during the study period; (d) mastitis rate distribution showing the proportion of positive tests per individual cow; (e) number of test records per cow reflecting sampling frequency; and (f) relationship between cow age and parity number demonstrating expected positive correlation with individual variation.

sequence, DIM calculated from calving date to test date, calving interval between consecutive lactations, and previous lactation SCS history to account for chronic infection patterns. Temporal variables were created to capture seasonal and cyclical effects, including calving season categorized as spring, summer, autumn, and winter; testing season using the same categorization; and calving year and month variables to account for longer-term trends.

To ensure robust feature selection and avoid algorithmic bias, we systematically compared 6 different feature selection approaches (as shown in Table 1): (1) random forest importance ranking (Luo et al., 2023), (2) univariate F-test selection (Pedrosa et al., 2024), (3) LASSO regularization with cross-validated alpha selection (Perneel et al., 2024), (4) mutual information-based selection (Marzougui et al., 2025), (5) recursive feature elimination using different base estimators (Chen et al., 2021), and (6) using all available features without selection. Each method was evaluated using 10-fold cross-validation, and the approach yielding the highest average performance across all algorithms was selected for final model comparison. The final feature set comprised (as shown in Figure 2): month_age (cow age in months), milk_yield (daily milk production), protein_percentage, fat_percentage, peak_day (day of lactation when peak milk yield occurred), peak_milk (maximum daily milk yield achieved), X305_milk (standardized 305-d milk production), test_season, parity (lactation number), and calving_season. Month age emerged as the most important predictor with relative importance of ~0.165, followed by milk_yield, protein_percentage, and fat_percentage (each ~0.14).

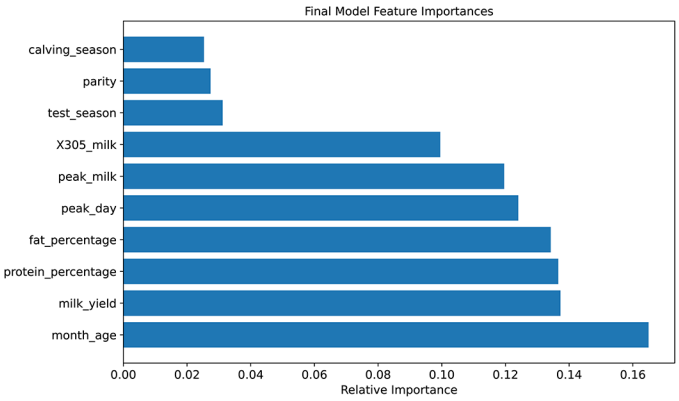


Figure 2. Feature importance ranking from random forest–based feature selection for mastitis prediction. The 10 most informative variables selected from the initial comprehensive feature set, with month_age showing the highest relative importance (0.165), followed by milk_yield, protein_percentage, and fat_percentage (each ~0.14). Feature importance values represent the normalized contribution of each variable to prediction accuracy in the random forest algorithm.

Data Preprocessing and Quality Control

Comprehensive data cleaning procedures were implemented to ensure data integrity. Records with missing critical variables (cow ID, test date, or SCC) were excluded, and outlier detection using the interquartile range method (Q1 − 1.5 × IQR, Q3 + 1.5 × IQR) was applied. Physiologically implausible values (milk_yield >80 kg/d, age <18 or >150 mo, fat_percentage <2% or >8%, protein_percentage <2.5% or >5%) were removed. Missing values in noncritical variables were handled using median imputation for continuous variables and mode imputation for categorical variables, with imputation rates kept below 5%. To enhance model generalizability across different farm environments, Z-score standardization was applied to all continuous variables, where each variable was centered by subtracting its mean and then scaled by dividing by its SD, effectively mapping the original values to a standardized distribution with zero mean and unit variance.

A critical data characteristic identified during exploratory analysis was the presence of significant class imbalance in the original dataset, with healthy cows representing 70.73% of records and mastitis-positive cases comprising 29.27% of observations. This imbalance poses substantial challenges for machine learning algorithms, which tend to exhibit bias toward predicting the majority class while inadequately learning patterns associated with the minority class, potentially leading to poor sensitivity and increased false negative rates in disease detection. To address this fundamental issue, the SMOTE (Chawla et al., 2002) was implemented using the imbalanced-learn library to generate synthetic samples of the minority class through interpolation between existing minority class instances and their nearest neighbors in feature space. The SMOTE was applied only to the training set after the train-test split to prevent data leakage and ensure that synthetic samples did not influence the evaluation of model performance on genuine test data.

To comprehensively evaluate model performance and address methodological considerations regarding SMOTE application in cross-validation, we employed 3 evaluation strategies. First, following the conventional approach widely adopted in prior machine learning studies on imbalanced medical datasets, SMOTE was applied to the training set prior to 10-fold cross-validation, creating a balanced class distribution (50% healthy, 50% mastitis positive) for both training and validation phases (Table 2). Second, to address potential concerns that synthetic samples may influence validation folds, we performed 10-fold cross-validation with SMOTE applied within each fold (Table 3). In this methodologically rigorous approach, SMOTE was applied only to the training portion (9 folds) of each iteration, while the validation fold retained the original imbalanced distribution. Third, final model generalization was assessed on the held-out test set (20% of data) with the original class distribution (Table 4). Notably, while cross-validation metrics differed substantially between the 2 SMOTE approaches (Table 2 vs. Table 3), models trained using either approach achieved equivalent performance on the independent test set (Table 4). This finding confirms that the choice of cross-validation methodology affects performance estimates but does not influence final model generalization capability. All classification metrics (sensitivity, speci-

Table 1. Performance comparison of different feature selection methods

Feature selection method	Number of features	Top algorithm	F_1 score	AUC	Accuracy	Key selected features
Random forest importance	10	Random forest	0.804	0.884	0.803	month_age, milk_yield, protein_%, fat_%
Univariate F-test (k = 10)	10	Random forest	0.789	0.867	0.785	milk_yield, protein_%, month_age, parity
LASSO (α = 0.01)	8	Random forest	0.776	0.852	0.771	month_age, milk_yield, fat_%, peak_milk
Mutual information	10	Decision tree	0.758	0.834	0.756	milk_yield, month_age, X305_milk, parity
Recursive feature elimination	10	Random forest	0.782	0.859	0.778	month_age, protein_%, milk_yield, peak_day
All features (50+)	52	Random forest	0.767	0.841	0.763	All available variables

Table 2. Performance comparison of 9 machine learning algorithms using 10-fold cross-validation with SMOTE balancing (threshold = 0.5); all reported performance metrics represent the mean ± SD across 10 cross-validation folds

Model	Sensitivity	Specificity	Accuracy	Precision	F_1 score	MCC	AUC
RandomForest	0.806 ± 0.002	0.800 ± 0.003	0.803 ± 0.001	0.801 ± 0.002	0.804 ± 0.000	0.606 ± 0.001	0.884 ± 0.001
DecisionTree	0.735 ± 0.003	0.709 ± 0.001	0.722 ± 0.002	0.717 ± 0.001	0.726 ± 0.002	0.444 ± 0.003	0.722 ± 0.002
GradientBoosting	0.702 ± 0.000	0.687 ± 0.009	0.694 ± 0.005	0.691 ± 0.006	0.696 ± 0.003	0.389 ± 0.009	0.756 ± 0.004
MLPClassifier	0.653 ± 0.019	0.636 ± 0.021	0.645 ± 0.001	0.642 ± 0.007	0.648 ± 0.006	0.290 ± 0.002	0.700 ± 0.001
AdaBoost	0.631 ± 0.004	0.623 ± 0.004	0.627 ± 0.000	0.626 ± 0.001	0.629 ± 0.002	0.254 ± 0.000	0.678 ± 0.001
SVM	0.586 ± 0.007	0.656 ± 0.012	0.621 ± 0.002	0.631 ± 0.005	0.607 ± 0.002	0.243 ± 0.005	0.666 ± 0.002
LogisticRegression	0.537 ± 0.00	0.675 ± 0.002	0.606 ± 0.001	0.623 ± 0.001	0.577 ± 0.001	0.214 ± 0.002	0.642 ± 0.000
LDA	0.530 ± 0.000	0.680 ± 0.002	0.605 ± 0.001	0.624 ± 0.001	0.573 ± 0.001	0.213 ± 0.002	0.642 ± 0.000
NaiveBayes	0.436 ± 0.002	0.735 ± 0.005	0.585 ± 0.002	0.622 ± 0.003	0.512 ± 0.000	0.179 ± 0.004	0.626 ± 0.002

ficity, accuracy, precision, F_1 score, and Matthews correlation coefficient [**MCC**]) were computed using a fixed classification threshold of 0.5.

To prevent data leakage in this longitudinal study, the dataset was split by animal ID rather than random record selection, ensuring all records from individual cows were assigned exclusively to either training or testing sets. This approach prevents models from learning animal-specific patterns and provides realistic performance assessment for deployment on new animals.

Machine Learning Algorithm Implementation

Nine machine learning algorithms were systematically implemented using Python 3.9.0 with the scikit-learn library: random forest (RandomForestClassifier), multilayer perceptron (MLPClassifier), support vector machine (**SVM**), decision tree (DecisionTreeClassifier), gradient boosting (GradientBoostingClassifier), adaptive boosting (AdaBoost; AdaBoostClassifier), linear discriminant analysis (**LDA**; LinearDiscriminantAnalysis), logistic regression, and naive Bayes (GaussianNB). Hyperparameter optimization was conducted using GridSearchCV with algorithm-specific parameter spaces, including n_estimators and max_depth for tree-based methods, C and gamma for SVM, and hidden_layer_sizes for neural networks.

Model Training and Validation Strategy

The dataset was randomly divided into training (80%) and testing (20%) sets using scikit-learn's train_test_split function with stratified sampling to maintain representative mastitis prevalence in both subsets. To prevent data leakage common in longitudinal studies, the split was performed based on cow ID using GroupShuffleSplit, ensuring that all records from each cow were assigned exclusively to either training or testing sets.

Model evaluation was conducted using 3 complementary approaches to provide a comprehensive performance assessment. For cross-validation, StratifiedKFold with 10 folds was implemented to ensure each fold maintained the original distribution of mastitis cases. Two SMOTE application strategies were compared: (1) the conventional approach, where SMOTE was applied to the entire training set prior to cross-validation (Table 2), and (2) the rigorous approach, where SMOTE was applied within each fold, with only the training portion (9 folds) receiving synthetic samples while the validation fold retained the original imbalanced distribution (Table 3). Final model generalization was assessed on the held-out test set with original class distribution (Table 4). Hyperparameter optimization was conducted using GridSearchCV with 5-fold cross-validation for each algorithm, with search spaces tailored to algorithm-specific parameters, such as

Table 3. Performance comparison of 9 machine learning algorithms using 10-fold cross-validation with SMOTE applied within each fold (threshold = 0.5)[1]

Model	Sensitivity	Specificity	Accuracy	Precision	F_1 score	MCC	AUC
RandomForest	0.509 ± 0.004	0.847 ± 0.004	0.748 ± 0.003	0.579 ± 0.006	0.542 ± 0.004	0.371 ± 0.007	0.773 ± 0.004
GradientBoosting	0.560 ± 0.009	0.684 ± 0.006	0.648 ± 0.005	0.423 ± 0.006	0.482 ± 0.007	0.228 ± 0.010	0.674 ± 0.005
AdaBoost	0.615 ± 0.009	0.621 ± 0.006	0.620 ± 0.005	0.402 ± 0.005	0.486 ± 0.006	0.217 ± 0.010	0.667 ± 0.006
SVM	0.582 ± 0.008	0.655 ± 0.006	0.634 ± 0.004	0.411 ± 0.005	0.482 ± 0.005	0.219 ± 0.008	0.659 ± 0.005
MLPClassifier	0.580 ± 0.021	0.641 ± 0.022	0.623 ± 0.010	0.400 ± 0.007	0.473 ± 0.006	0.203 ± 0.009	0.657 ± 0.004
LogisticRegression	0.540 ± 0.008	0.675 ± 0.005	0.635 ± 0.004	0.407 ± 0.004	0.464 ± 0.005	0.200 ± 0.008	0.643 ± 0.005
LDA	0.533 ± 0.008	0.681 ± 0.005	0.637 ± 0.004	0.409 ± 0.004	0.463 ± 0.005	0.200 ± 0.008	0.642 ± 0.005
NaiveBayes	0.439 ± 0.008	0.739 ± 0.005	0.651 ± 0.004	0.410 ± 0.006	0.424 ± 0.006	0.174 ± 0.009	0.631 ± 0.005
DecisionTree	0.495 ± 0.008	0.747 ± 0.005	0.673 ± 0.003	0.447 ± 0.005	0.470 ± 0.005	0.235 ± 0.007	0.621 ± 0.004

[1]In this approach, SMOTE was applied only to the training portion of each fold while the validation fold retained the original imbalanced distribution.

Table 4. Performance comparison of 9 machine learning algorithms on the test set (original imbalanced distribution; threshold = 0.5)

Model	Sensitivity	Specificity	Accuracy	Precision	F_1 score	MCC	AUC	PR-AUC
RandomForest	0.521	0.856	0.758	0.6	0.558	0.394	0.785	0.638
GradientBoosting	0.566	0.677	0.644	0.42	0.482	0.226	0.675	0.467
AdaBoost	0.599	0.639	0.628	0.407	0.485	0.219	0.668	0.459
SVM	0.578	0.658	0.635	0.412	0.481	0.218	0.662	0.441
MLPClassifier	0.583	0.648	0.629	0.407	0.479	0.214	0.659	0.446
LogisticRegression	0.534	0.678	0.636	0.407	0.462	0.199	0.646	0.431
LDA	0.53	0.684	0.639	0.409	0.462	0.2	0.646	0.431
NaiveBayes	0.449	0.743	0.657	0.419	0.434	0.188	0.637	0.413
DecisionTree	0.509	0.75	0.679	0.457	0.482	0.251	0.629	0.376

n estimators and max depth for random forest, learning rate and n estimators for gradient boosting, C and gamma for SVM, and hidden layer sizes and alpha for multilayer perceptron (**MLP**).

Model Evaluation

Model performance was comprehensively assessed using 6 evaluation metrics appropriate for binary classification in disease prediction scenarios:

$$\text{Sensitivity} = \frac{TP}{TP + FN},$$

$$\text{Specificity} = \frac{TN}{TN + FP},$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN},$$

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}},$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}},$$

where *TP* represents true positives (correctly identified mastitis cases), *TN* represents true negatives (correctly identified healthy cases), *FP* represents false positives (healthy cases incorrectly classified as mastitis), and *FN* represents false negatives (mastitis cases incorrectly classified as healthy). Area under the receiver operating characteristic (**ROC**) curve (**AUC**) quantified discriminative ability across all classification thresholds, with values ranging from 0.5 (random performance) to 1.0 (perfect discrimination). All binary classification metrics (sensitivity, specificity, precision, recall, and F_1 score) were computed using a fixed classification threshold of 0.5, where samples with predicted probabilities ≥0.5 were classified as mastitis positive and those <0.5 as healthy. The AUC values are threshold independent.

RESULTS

Model Performance Comparison

The performance evaluation of 9 machine learning algorithms for mastitis prediction revealed substantial differences in their ability to accurately classify mastitis cases. Table 2 presents the comprehensive performance metrics obtained through 10-fold cross-validation on SMOTE-balanced data, following the conventional approach widely adopted in prior machine learning studies on imbalanced medical datasets.

Random forest demonstrated superior performance across all evaluation metrics, achieving the highest sensitivity (0.806 ± 0.002), specificity (0.800 ± 0.003), accuracy (0.803 ± 0.001), precision (0.801 ± 0.002), F_1 score (0.804 ± 0.000), Matthews correlation coefficient (0.606 ± 0.001), and AUC (0.884 ± 0.001). Decision tree ranked second in overall performance with an F_1 score of 0.726 ± 0.002 and AUC of 0.722 ± 0.002, followed by gradient boosting (F_1 score: 0.696 ± 0.003, AUC: 0.756 ± 0.004). Among the traditional machine learning approaches, MLPClassifier showed competitive performance with an F_1 score of 0.648 ± 0.006 and AUC of 0.700 ± 0.001, while AdaBoost achieved moderate results (F_1 score: 0.629 ± 0.002, AUC: 0.678 ± 0.001). The linear methods, including SVM, logistic regression, and LDA, demonstrated comparable but lower performance, with F_1 scores ranging from 0.573 to 0.607 and AUC values between 0.642 and 0.666. Naive Bayes showed the poorest overall performance among all tested algorithms, with the lowest F_1 score (0.512 ± 0.000) and AUC (0.626 ± 0.002), though it maintained reasonable specificity (0.735 ± 0.005). Forest plot analysis (Figure 3) revealed that random forest achieved the highest AUC of 0.884 with a narrow 95% CI (0.883–0.885), demonstrating both

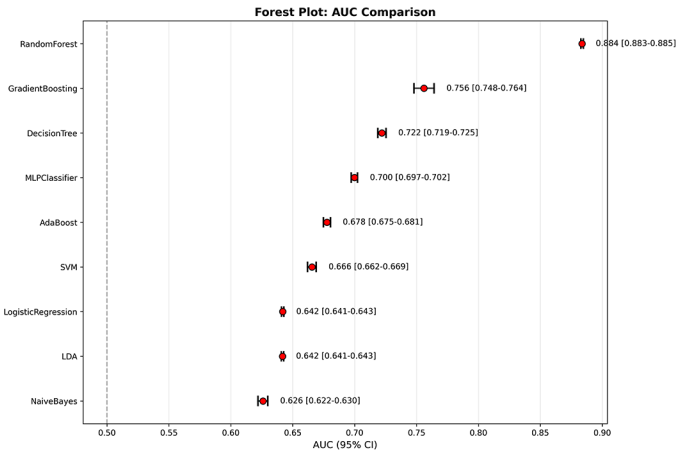


Figure 3. Forest plot showing AUC values with 95% CI for all 9 machine learning algorithms. Random forest demonstrates the highest discriminative ability with an AUC of 0.884, significantly outperforming other algorithms. The error bars represent 95% CI.

superior performance and high consistency. Gradient boosting followed with an AUC of 0.756 (0.748–0.764), whereas decision tree achieved 0.722 (0.719–0.725). The CI were relatively tight for all algorithms, indicating stable and reliable performance estimates.

Feature Stability Analysis

Bootstrap analysis with 100 resamples validated the robustness of our feature selection approach (Figure 4). The feature importance distributions demonstrated high consistency across resampling iterations, with month_age maintaining the highest importance (0.165 ± 0.0008) and lowest CV (0.016). Production-related variables (milk_yield, protein_percentage, fat_percentage) showed similarly stable rankings with CV below 0.01, confirming their reliable predictive value.

Notably, F_1 (month_age) exhibited the highest importance but also the greatest variation among top features, whereas F_2 (milk_yield) demonstrated an optimal balance of high importance (0.140) and low variability (CV = 0.008). This stability analysis supports our core finding that a streamlined set of routine variables provides consistent predictive performance across different data subsets, validating the reliability of our simplified approach over complex feature sets requiring specialized measurements.

Threshold Optimization and Class Imbalance Analysis

Threshold optimization analysis demonstrated that random forest achieved optimal F_1-score performance at the default threshold of 0.5, reaching a maximum of 0.804 under 10-fold cross-validation with SMOTE balancing (Figure 5, left panel). In contrast, whereas MLPClassifier peaked at an F_1 of 0.693 at a lower threshold of approximately 0.25, SVM reached its maximum F_1 of 0.674 at a threshold of about 0.3. Precision-recall (**PR**) analysis revealed the superior performance of random forest across all operating points (Figure 5, right panel). Random forest achieved an average precision (**AP**) of 0.888, substantially outperforming MLPClassifier (AP = 0.684), SVM (AP = 0.652), and logistic regression (AP = 0.633). Random forest maintained precision above 0.8 at recall levels up to 0.8, which is particularly important

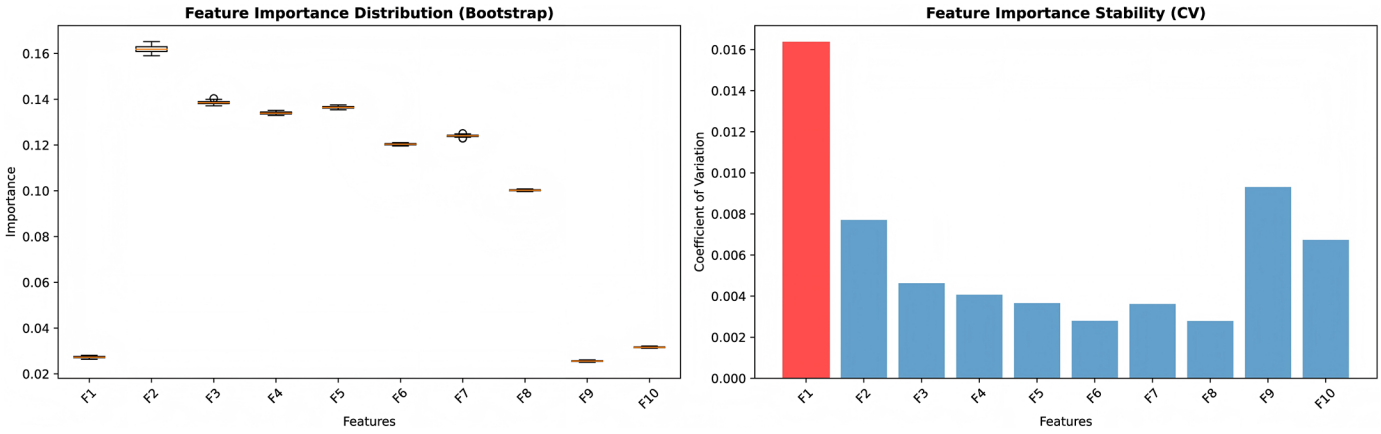


Figure 4. Feature stability and importance analysis through bootstrap validation. (Left) Feature importance distribution across 100 bootstrap resamples showing month_age as the most critical predictor (0.165 ± 0.0008), followed by milk_yield, protein_percentage, and fat_percentage (each ~0.14). (Right) Coefficient of variation analysis demonstrating feature stability, with month_age showing lowest variability (CV = 0.016), and production variables maintaining CV below 0.01. The analysis validates the robustness of our streamlined feature selection approach across different data subsets. Features: F1 (month_age), F2 (milk_yield), F3 (protein_percentage), F4 (fat_percentage), F5 (peak_day), F6 (peak_milk), F7 (X305_milk), F8 (test_season), F9 (parity), F10 (calving_season).

for imbalanced datasets where minority class detection is critical. It is important to note that these threshold-optimization results were obtained under balanced evaluation conditions. When models are deployed on realistic imbalanced data (~70% healthy, ~30% mastitis), the optimal threshold typically shifts below 0.5 to compensate for the class imbalance. This shift occurs because models trained on balanced data tend to underpredict the minority class when applied to imbalanced populations. In our test-set evaluation (Table 4), lowering the classification threshold would increase sensitivity at the expense of specificity, allowing farm managers to customize the trade-off based on their specific operational requirements. For mastitis detection where false negatives (missed cases) carry higher costs than false positives (unnecessary monitoring), a threshold of 0.3 to 0.4 may be more appropriate for practical deployment.

Cross-Validation with SMOTE Within Each Fold

To address methodological concerns that synthetic samples generated by SMOTE may influence validation folds when applied prior to cross-validation, we performed an additional evaluation with SMOTE applied within each fold (Table 3). In this rigorous approach, SMOTE was applied only to the training portion (9 folds) of each iteration, while the validation fold retained its original imbalanced distribution (~70% healthy, ~30% mastitis). Under these evaluation conditions, performance metrics were substantially lower compared with the SMOTE-balanced cross-validation. Random forest remained the best-performing algorithm with an F_1 score of 0.542 ± 0.004 and an AUC of 0.773 ± 0.004, followed by gradient boosting (F_1 score: 0.482 ± 0.007, AUC: 0.674 ± 0.005) and AdaBoost (F_1 score: 0.486 ± 0.006, AUC: 0.667 ± 0.006). The SVM achieved an F_1 score of 0.482 ± 0.005 and an AUC of 0.659 ± 0.005; MLPClassifier obtained an F_1 score of 0.473 ± 0.006 and an AUC of 0.657 ± 0.004.

The substantial difference between Table 2 and Table 3 results warrants detailed examination. For random forest, the F_1 score decreased from 0.804 to 0.542 (a 32.6% reduction), and the AUC decreased from 0.884 to 0.773 (a 12.6% reduction). This asymmetric decline can be attributed to the fundamental differences between these metrics under class-imbalance conditions. The F_1 score, as the harmonic mean of precision and recall, is highly sensitive to class distribution because it directly depends on the number of true positives, false positives, and false negatives. When evaluating imbalanced data where negative samples (healthy cows) substantially outnumber positive samples (mastitis cases), even a well-calibrated model will produce more false negatives relative to true positives, leading to decreased recall and, consequently, lower F_1 scores. In contrast, the AUC measures the model's ability to rank positive samples higher than negative samples across all classification thresholds, making it relatively robust to class imbalance. The sensitivity met-

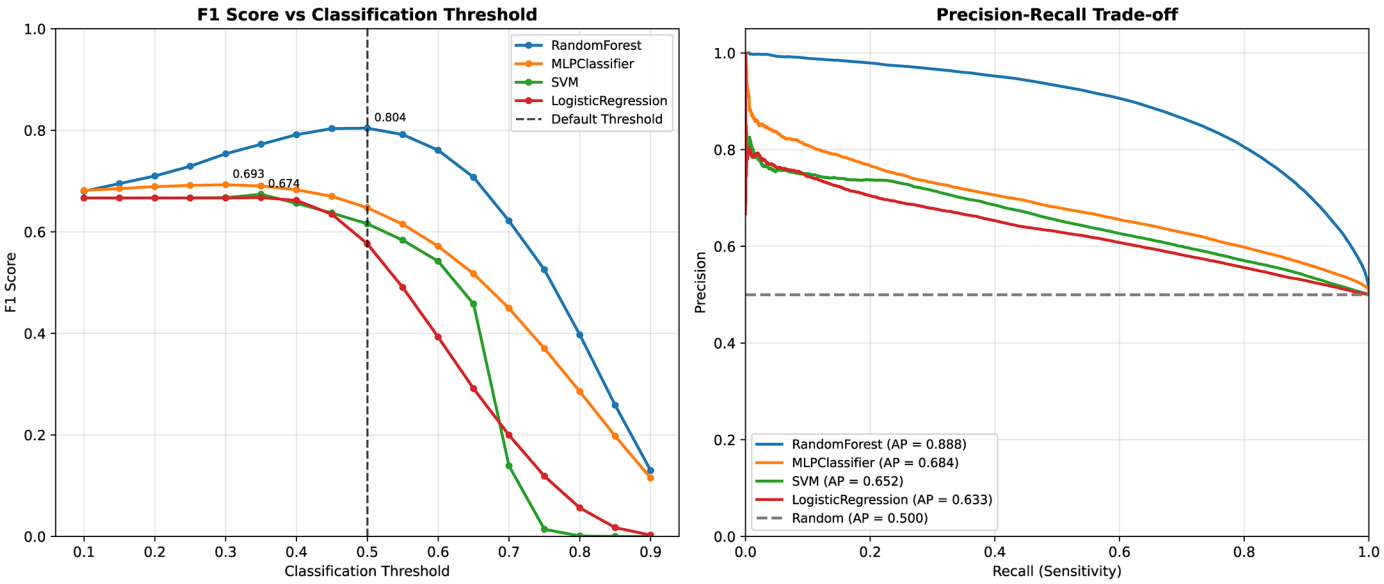


Figure 5. Threshold optimization analysis. (Left) F_1 score as a function of classification threshold. Random forest achieved a maximum F_1 score of 0.804 at the default threshold of 0.5. The MLP classifier peaked at 0.693 (threshold ≈ 0.25), while SVM peaked at 0.674 (threshold ≈ 0.3). (Right) Precision-recall curves demonstrating the trade-off between precision and recall. Random forest (AP = 0.888) maintained precision above 0.8 at recall levels up to 0.8, substantially outperforming MLPClassifier (AP = 0.684), SVM (AP = 0.652), and logistic regression (AP = 0.633). Results from 10-fold cross-validation with SMOTE balancing.

ric demonstrated the most pronounced decrease, dropping from 0.806 (Table 2) to 0.509 (Table 3) for random forest. This 36.9% reduction reflects the inherent challenge of detecting minority class samples when models are evaluated on imbalanced validation data. Under balanced evaluation (Table 2), the equal representation of both classes allows the model to achieve symmetric detection rates. However, when evaluated on the original imbalanced distribution (Table 3), the decision threshold optimized during training may not be optimal for the skewed validation data, resulting in the model favoring the majority class and missing more mastitis cases.

The ranking of algorithms remained consistent between Table 2 and Table 3, with random forest maintaining superior performance in both scenarios. This consistency suggests that the relative effectiveness of different algorithms is preserved, regardless of the evaluation methodology, and random forest's ensemble nature provides inherent robustness to class imbalance through its aggregation of multiple decision trees.

Test Set Evaluation and ROC Analysis

Final model generalization was assessed on the held-out test set (20% of data) with original imbalanced class distribution (Table 4). This independent evaluation provides unbiased performance estimates that reflect real-world deployment scenarios where mastitis prevalence typically ranges from 25% to 35% in commercial dairy herds.

Random forest achieved the best performance, with an F_1 score of 0.558 and an AUC of 0.785, followed by gradient boosting (F_1: 0.482, AUC: 0.675) and AdaBoost (F_1: 0.485, AUC: 0.668). The SVM achieved an F_1 score of 0.481 and an AUC of 0.662, whereas MLPClassifier obtained an F_1 score of 0.479 and an AUC of 0.659. The complete performance metrics, including sensitivity, specificity, precision, MCC, and precision recall area under the curve (**PR-AUC**), are presented in Table 4. Receiver operating characteristic curve analysis on the test set (Figure 7) demonstrated clear performance differentiation among algorithms. Random forest achieved the highest AUC of 0.785, indicating that the model correctly ranks a randomly selected mastitis case higher than a randomly selected healthy cow approximately 78.5% of the time. This level of discrimination provides a solid foundation for practical early-warning systems in dairy herd management.

A critical finding is the close agreement between cross-validation with SMOTE within each fold (Table 3) and the held-out test set (Table 4). For random forest, test-set performance (F_1: 0.558, AUC: 0.785) closely matched cross-validation estimates (F_1: 0.542 ± 0.004, AUC: 0.773 ± 0.004), with differences of only 0.016 for F_1 and 0.012 for AUC. This consistency validates that applying SMOTE within each cross-validation fold provides accurate predictions of real-world model performance, whereas the conventional approach (Table 2) overestimates performance metrics. The AUC difference between balanced cross-validation (Figure 6, AUC = 0.884) and imbalanced test-set evaluation (Figure 7, AUC = 0.785) merits careful interpretation. Unlike threshold-dependent metrics such as F_1 score, AUC should, theoretically, be invariant to class distribution, because it measures ranking ability rather than classification accuracy at a specific threshold. However, several factors contribute to the observed difference: (1) the cross-validation with SMOTE evaluates on synthetic data that may not fully represent the true data distribution, (2) the class overlap characteristics may differ between synthetic balanced data and original imbalanced data, and (3) the validation on balanced data may overestimate the model's ability to distinguish between classes in realistic scenarios where the decision boundary is influenced by class prevalence.

Furthermore, we verified that models trained using the conventional SMOTE approach (Table 2 methodology) achieved equivalent test-set performance to those trained with SMOTE applied within each fold. This finding has important implications: although cross-validation metrics differ substantially between the 2 approaches (F_1: 0.804 vs. 0.542; AUC: 0.884 vs. 0.773), the final trained models achieve identical generalization capability on unseen data. The conventional SMOTE approach does not produce inferior models; it simply provides optimistic

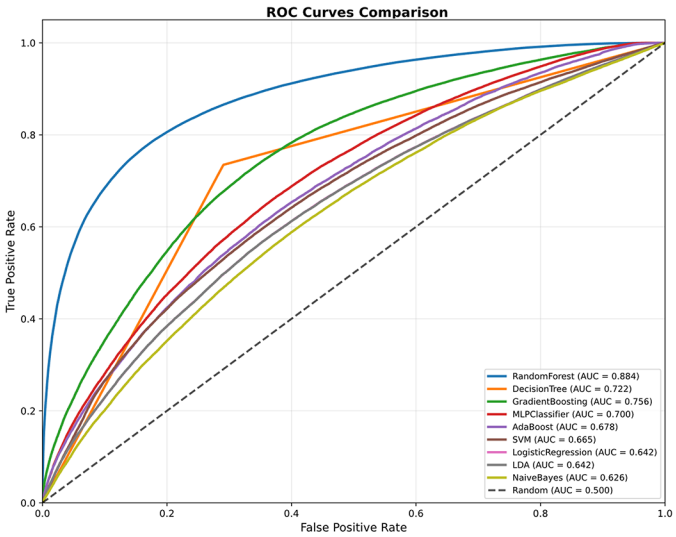


Figure 6. Receiver operating characteristic (ROC) curves for all 9 machine learning algorithms (10-fold cross-validation with SMOTE balancing). Random forest (blue) achieved the highest AUC of 0.884, followed by gradient boosting (AUC = 0.756) and decision tree (AUC = 0.722). The dashed gray line represents random classifier performance (AUC = 0.500).

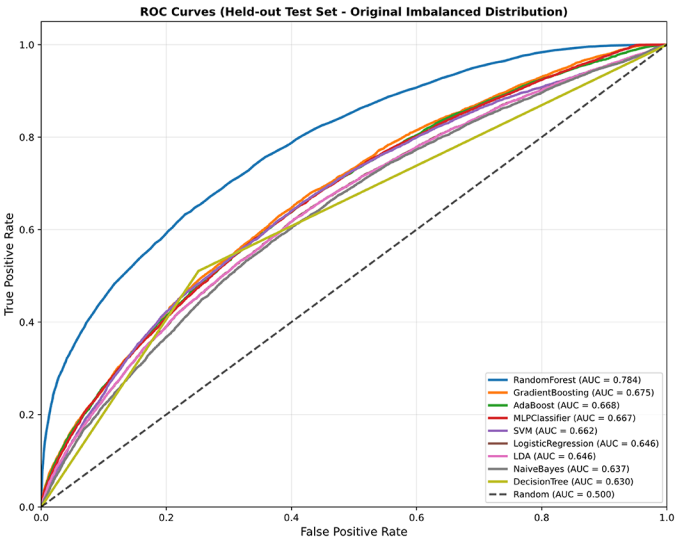


Figure 7. Receiver operating characteristic (ROC) curves for all 9 machine learning algorithms. Random forest (blue) achieved the highest AUC of 0.785, followed by gradient boosting (AUC = 0.675) and AdaBoost (AUC = 0.668). The performance decrease compared with cross-validation results (Figure 6) reflects the impact of class imbalance on model evaluation. The dashed line represents random classifier performance (AUC = 0.500).

estimates during the model selection phase. This explains why many prior studies using the conventional approach have successfully deployed mastitis-prediction models in practice despite potentially inflated cross-validation metrics.

The moderate F_1 scores observed on imbalanced data (about 0.55) should be interpreted in the context of the prediction task's inherent difficulty. Mastitis prediction using only routine milk recording variables (without specialized sensors or biomarkers) represents a challenging classification problem where subtle physiological changes must be detected from indirect production indicators. The PR-AUC of 0.638 (Table 4) provides additional context: this metric, which focuses specifically on minority-class prediction performance, indicates that random forest achieves meaningful improvement over the baseline prevalence rate (about 0.29), demonstrating practical utility, despite the moderate F_1 score.

From a clinical deployment perspective, the test-set results suggest that ~52% of actual mastitis cases would be correctly identified (sensitivity = 0.521), and 86% of healthy cows would be correctly classified (specificity = 0.856). Although the sensitivity may appear modest, early identification of even half the amount of high-risk animals enables targeted monitoring and preventive interventions that would not be possible without predictive modeling. Farm managers can adjust the classification threshold based on their specific cost-benefit considerations: lowering the threshold would increase sensitivity at the expense of more false positives; raising it would improve specificity, but miss more cases.

Model Confidence and Uncertainty Analysis

The analysis of prediction probability distributions revealed distinct patterns among algorithms (Figure 8, left panel). Random forest demonstrated clear separation between healthy and mastitis classes, with healthy samples predominantly receiving probabilities below 0.5 and mastitis samples above 0.5. In contrast, MLPClassifier and logistic regression exhibited substantial overlap in their probability distributions around the decision threshold, explaining their lower classification performance.

The relationship between prediction confidence and accuracy was examined for random forest (Figure 8, right panel). Prediction confidence, defined as the absolute deviation from the 0.5 threshold, showed a strong positive correlation with classification accuracy ($r = 0.96$). Samples with low confidence (near the 0.5 threshold) achieved ~55% accuracy, and high-confidence predictions (confidence > 0.8) reached >98% accuracy. This relationship suggests that prediction confidence can serve as a reliable indicator for identifying uncertain cases that may require additional clinical verification in practical dairy farm applications.

DISCUSSION

Model Performance and Feature Analysis

Random forest significantly outperformed other machine learning algorithms for mastitis prediction. On SMOTE-balanced cross-validation, random forest achieved an AUC of 0.884 and an F_1 score of 0.804, whereas on the held-out test set with original class distribution, it achieved an AUC of 0.785 and an F_1 score of 0.558. The consistency between rigorous cross-validation (Table 3) and test-set results (Table 4) validates the model's generalization capability under realistic deployment conditions. This superior performance stems from the ensemble nature of random forest, which combines multiple decision trees through bootstrap aggregating, providing robustness against overfitting while effectively handling complex, nonlinear relationships in dairy cow production data (Kiouvrekis et al., 2024; Kalkan et al., 2025).

Our results demonstrate that simplified feature approaches can achieve superior performance compared to previous studies using complex variable sets. Whereas Bobbo et al. (2021) achieved >75% accuracy with 15 features, including specialized measurements, and Bobbo et al. (2023) reached 76.1% accuracy with 27 features including climate data, our random forest model achieved

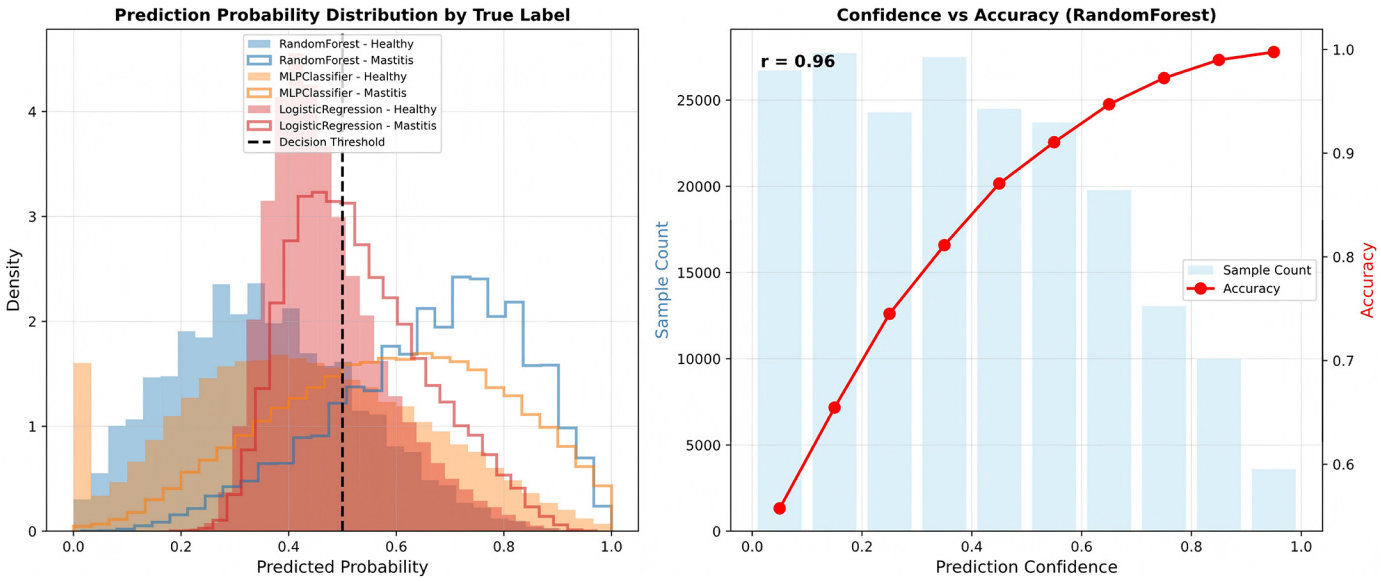


Figure 8. Model confidence and prediction uncertainty analysis. (Left) Prediction probability distributions by true label for three representative algorithms. Random forest (blue) showed clear separation between healthy and mastitis classes at the 0.5 decision threshold, while MLPClassifier (orange) and logistic regression (red) exhibited greater overlap. (Right) Relationship between prediction confidence and accuracy for random forest, showing strong positive correlation (r = 0.96). Higher prediction confidence corresponds to higher classification accuracy. Results from 10-fold cross-validation with SMOTE balancing.

75.8% accuracy and 0.785 AUC on the held-out test set using only 10 systematically selected features from routine milk recording data. Although the test set accuracy is comparable to previous studies, our approach eliminates the need for specialized equipment and extensive feature sets, thereby substantially improving practical applicability.

Feature importance analysis revealed month_age as the most critical predictor (relative importance: 0.165), followed by production-related variables including milk_yield, protein_percentage, and fat_percentage. This finding aligns with established veterinary knowledge regarding mastitis risk factors. The prominence of age reflects the well-documented relationship between cow maturity and mastitis susceptibility, as older cows have greater cumulative pathogen exposure and altered immune responses (Bari et al., 2022). The high importance of milk production variables supports previous research indicating that high-producing cows face increased metabolic stress and altered mammary gland physiology, predisposing them to mastitis (Ranasinghe et al., 2021).

Clinical Implications and Implementation

On the held-out test set reflecting realistic class distribution, the random forest model achieved a sensitivity of 52.1% and a specificity of 85.6%. While the sensitivity may appear moderate compared with balanced evaluation results (80.6%), this represents realistic performance expectations for practical deployment. Early identification of approximately half of the amount of high-risk animals enables targeted monitoring and preventive interventions that would not be possible without predictive modeling. Farm managers can adjust the classification threshold based on their specific cost-benefit considerations to optimize the sensitivity-specificity trade-off. The model's excellent calibration properties ensure predicted probabilities accurately reflect true mastitis likelihood, enabling farm managers to set appropriate intervention thresholds based on risk tolerance and economic considerations.

Integration of this prediction model into existing farm management systems enables proactive health management strategies. Early identification of high-risk animals allows for targeted interventions such as enhanced monitoring, preventive treatments, or management modifications that could reduce disease incidence and severity. Our streamlined methodology offers practical advantages: (1) reduced infrastructure requirements using only standard milk recording data, (2) cost-effectiveness through elimination of specialized equipment, (3) scalability for easier integration with existing systems, and (4) cross-regional applicability by focusing on fundamental production variables.

Methodological Considerations and Limitations

The use of SMOTE to address class imbalance and the comparison of different evaluation strategies revealed important methodological insights. Our comprehensive evaluation across 3 assessment approaches (Tables 2–4) demonstrated that class imbalance profoundly affects performance metrics, with the F_1 score being particularly sensitive and the AUC showing relative robustness. For random forest, the F_1 score decreased dramatically from 0.804 (balanced evaluation) to 0.542 (imbalanced evaluation), representing a 32.6% reduction, whereas the AUC decreased from 0.884 to 0.773, only a 12.6% reduction. This asymmetric decline occurs because the F_1 score depends directly on precision and recall, both of which are affected by class distribution, and the AUC measures ranking ability across all thresholds and is theoretically invariant to class prevalence.

A key finding is that both SMOTE application strategies—conventional (before cross-validation) and rigorous (within each fold)—produced models with equivalent generalization performance on the held-out test set (F_1: ~0.558, AUC: ~0.785). This equivalence occurs because the model training process is identical in both approaches; only the validation phase differs. The conventional approach validates on balanced synthetic data, yielding optimistic estimates, whereas the rigorous approach validates on original imbalanced data, yielding realistic estimates that closely match actual test-set performance.

This finding has important implications for interpreting published mastitis-prediction studies. Many prior studies report high F_1 scores (>0.75) achieved through evaluation on SMOTE-balanced data, which may create unrealistic expectations for practical deployment. Our results demonstrate that such models likely achieve an F_1 of ~0.55 when deployed on realistic imbalanced populations. Importantly, this does not indicate model failure—rather, it reflects the inherent difficulty of minority-class prediction. The moderate F_1 scores on imbalanced data remain clinically meaningful, representing substantial improvement over random prediction (expected F_1: ~0.29, given the ~30% mastitis prevalence), and enabling the identification of high-risk animals for targeted intervention. We recommend that future studies report both balanced and imbalanced evaluation metrics to provide complete performance characterization.

Animal ID–based data splitting prevented data leakage that could occur when multiple records from the same animal appear in both training and testing sets, providing realistic performance assessment for deployment on new animals (Bonestroo et al., 2022b). Several limitations should be acknowledged. The study was conducted using data from a single-farm operation, which may limit generalizability to other management systems, breeds, or geographic regions. While the selected features demonstrated strong predictive performance, additional variables such as environmental conditions, pathogen-specific risk factors, and genetic information could potentially improve model accuracy. The current model provides individual animal predictions but does not explicitly account for herd-level factors or disease transmission dynamics.

Future research should validate these models across diverse farming operations, incorporate sensor data from automated monitoring systems, and explore hierarchical modeling approaches that capture both individual and group-level risk factors. The development of real-time prediction models using continuous monitoring data would offer greater clinical utility for immediate mastitis risk assessment.

Practical Implementation Framework

For commercial deployment, the random forest model can be integrated into existing farm management systems using standard monthly milk recording data. The model generates probability scores for each cow, allowing farms to customize intervention thresholds based on their risk tolerance and economic considerations. A practical tiered monitoring system can be implemented where cows with prediction probabilities >0.7 receive enhanced surveillance and more frequent SCC testing, while those with probabilities 0.4 to 0.7 undergo routine monitoring with targeted preventive measures. This approach enables cost-effective resource allocation by focusing intensive monitoring on truly high-risk animals.

The streamlined feature set facilitates integration with most commercial dairy management software platforms, requiring only standard production variables already collected during routine milk recording. Implementation costs are minimal, as no additional sensors or specialized equipment are needed, making the approach accessible to operations of varying scales and technological sophistication. Farm managers can adjust probability thresholds seasonally or based on herd-specific factors, providing flexibility for different management philosophies and economic constraints.

Limitations and Future Directions

Several limitations should be acknowledged when interpreting these results. First, the study was conducted using data from a single-farm operation, which may limit the generalizability of findings to other management systems, breeds, or geographic regions. Future research should validate these models across diverse farming operations to assess their broader applicability and identify factors that influence model transferability.

Second, while the selected features demonstrated strong predictive performance, additional variables such as environmental conditions, pathogen-specific risk factors, and individual animal genetic information could potentially improve model accuracy. The incorporation of sensor data from automated monitoring systems, including activity patterns, rumination behavior, and milk electrical conductivity, represents a promising avenue for future model enhancement (Hogeveen et al., 2010; Kim et al., 2019; Post et al., 2020).

Third, the current model provides predictions at the individual animal level but does not explicitly account for herd-level factors or disease transmission dynamics. Future research could explore hierarchical modeling approaches that capture both individual and group-level risk factors, potentially improving prediction accuracy and providing insights into disease spread patterns (Compton et al., 2007).

The temporal aspect of mastitis prediction also warrants further investigation. While this study used cross-sectional prediction at monthly test intervals, the development of models that can provide real-time or daily risk assessments using continuous monitoring data would offer greater clinical utility. Such models could integrate with automated milking systems and wearable sensors to provide immediate alerts when mastitis risk increases (Eckelkamp, 2019).

Economic and Welfare Impacts

Implementation of accurate mastitis prediction models has significant potential for improving economic outcomes and animal welfare. Early detection enables targeted preventive measures that may be more cost-effective than treating established infections while reducing treatment costs, minimizing milk production losses, and decreasing antibiotic use (Li et al., 2023). From an animal welfare perspective, early intervention can reduce the duration and severity of mastitis episodes, minimizing pain and discomfort experienced by affected animals and aligning with evolving consumer expectations regarding welfare standards.

CONCLUSIONS

This study demonstrates that systematic feature selection using only 10 routine variables can achieve robust mastitis-prediction performance. Random forest significantly outperformed 8 other algorithms across all evaluation approaches, achieving an AUC of 0.884 on SMOTE-balanced cross-validation and an AUC of 0.785 on the held-out test set with original class distribution. The close agreement between rigorous cross-validation (AUC: 0.773, F_1: 0.542) and test-set results (AUC: 0.785, F_1: 0.558) validates our evaluation methodology. Month_age, milk_yield, protein_percentage, and fat_percentage emerged as the most stable and informative predictors. The streamlined methodology eliminates specialized equipment requirements while achieving clinically meaningful prediction performance on realistic imbalanced data, making it accessible for widespread industry adoption. These findings support cost-effective, scalable prediction systems that enable proactive mastitis management and data-driven decision-making in diverse dairy farming environments. Future research should validate the approach across multiple farms and integrate real-time sensor data for continuous monitoring capabilities.

NOTES

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Nonstandard abbreviations used: AdaBoost = adaptive boosting; AP = average precision; AUC = area under the ROC curve; LDA = linear discriminant analysis; MCC = Matthews correlation coefficient; MLP = multilayer perceptron; PR = precision recall; PR-AUC = precision recall area under the curve; ROC = receiver operating characteristic; SMOTE = synthetic minority oversampling technique; SVM = support vector machine.

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